Verum Interior Gauge (VIG): Version 7 Final

Comparative Critical-Growth Laws for Regular Black-Hole Interiors

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Abstract

Version 7 Final presents the polished comparative form of the Verum Interior Gauge (VIG) program. The original motivating question was whether black-hole interiors admit a universal saturation clock of the form

$$\tau_{\text{sat}} \sim CMe^{S}.$$

The geometry-first program led to a sharper and better-supported object: for each regular interior model studied so far, the near-critical maximal-slice family is governed by a model-dependent critical growth constant

$$C_{\text{crit}}(M, u) = 4\pi M^2\, \mathcal{S}(u),$$

where u is a normalized deformation parameter. In this paper, the Hayward and Bardeen interiors are treated in a uniform formalism, each with an exact dimensionless law, a Schwarzschild anchor, a weak-deformation asymptotic, and a near-extremal asymptotic. The numerical family scans show the same structural law in both cases:

$$\text{tail mean} \approx C_{\text{crit}}.$$

The universality is therefore structural rather than exact-functional. Hayward and Bardeen belong to the same qualitative maximal-slice growth class, but they do not share the same deformation curve. In the full-edge overlap now available from the master driver, Bardeen remains below Hayward throughout and is approximately 16.3% lower in the normalized critical-growth law at the top of that overlap. The paper is tied explicitly to the source CSV products, the generating codes, and the master reproducibility driver.

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1 Introduction and motivation

The starting point of this project was a broad physical intuition: if black-hole interiors are not merely passive continuations of exterior geometry but dynamical regions with their own large-scale structure, then one should be able to describe that structure with observables that scale cleanly across models. The original framing emphasized a possible *Universal Clock*, inspired by the heuristic form

$$\tau_{\mathrm{sat}} \sim CMe^{S}.$$

That conjectural program remains interesting, but it is no longer the center of gravity of the work.

The geometry-first path produced a more immediate and more rigorously testable object. For spherically symmetric maximal slices in regular black-hole interiors, one finds a distinguished critical

constant $C_{\rm crit}$ controlling the near-critical family. In the present framework, the operative law is not an entropy-weighted saturation statement, but a comparative maximal-slice growth law,

$$V(v) \sim C_{\rm crit}^{(\rm model)}(M,u)\, v,$$

where M is the black-hole mass and u is a dimensionless deformation parameter. The entire project now revolves around three linked goals:

1. derive exact expressions for $C_{\rm crit}$ in concrete regular metrics,
2. determine the weak-deformation and near-extremal asymptotics,
3. establish which aspects of the resulting growth law are universal and which are model-specific.

The two regular models developed here are the Hayward metric and the Bardeen metric. The first is organized naturally by $\lambda = \ell/M$, where ℓ sets the Hayward core scale. The second is organized by $\beta = g/M$, where g is the Bardeen magnetic-charge scale. Both reduce to Schwarzschild when the deformation vanishes. Both admit extremal endpoints. And both now have exact analytic $C_{\rm crit}$-laws, numerical atlases, and verified asymptotics.

The vision of the VIG program is to turn regular black-hole interiors into a reproducible comparative cartography. In that spirit, this final version is tied directly to explicit code and exported data products: the Hayward unified framework, the Bardeen unified framework, the Bardeen dense edge scan, the full-edge Hayward–Bardeen comparison, and a master driver that aggregates the whole pipeline.

2 Context and prior work

The large-interior perspective for black holes was sharpened by Christodoulou and Rovelli, who showed that the maximal interior volume of a Schwarzschild black hole grows linearly with advanced time at the rate $3\sqrt{3}\,\pi M^2$ [1]. Bengtsson and Jakobsson clarified geometric aspects of these large interiors and emphasized how nontrivial the interior notion of volume becomes even in classical general relativity [2]. Subsequent work explored persistence of large interiors, thermodynamic questions, and the fate of such growth in modified scenarios [7, 8].

On the regular-black-hole side, Hayward proposed a nonsingular evaporating black-hole model that replaces the classical singularity by a finite core [3]. Bardeen's earlier regular black-hole construction remains historically foundational and has served as a template for many later nonsingular geometries [4]. Modern discussions of nonsingular or geodesically complete black holes, including the status of regular cores and interior structure, may be found in [5, 6].

At the level of interior growth and late-time structure, exact results in lower-dimensional gravity also motivate caution about extrapolating from classical growth to true saturation. In particular, Iliesiu, Mezei, and Sárosi showed that the volume of the black-hole interior in a solvable two-dimensional setting has a much richer late-time structure than a naive perpetual linear law would suggest [9]. That kind of result keeps the original clock motivation alive, but it does not substitute for a fully four-dimensional geometric derivation.

The present work therefore stays disciplined: it does not claim a derivation of an entropy-weighted clock. Instead, it proves and verifies comparative geometric growth laws for regular interiors, using code and data products that can be rerun directly.

3 Generic VIG formalism

3.1 Metric and graph gauge

Work in ingoing Eddington–Finkelstein coordinates,

$$ds^2 = -f(r)\,dv^2 + 2\,dv\,dr + r^2 d\Omega^2. \tag{1}$$

A spherically symmetric spacelike slice is written as a graph $v = v(r)$. Its induced radial coefficient is

$$\gamma_{rr} = -f(r)\,v'(r)^2 + 2v'(r), \tag{2}$$

and the corresponding volume functional is

$$V = \int 4\pi r^2 \sqrt{-f(r)\,v'(r)^2 + 2v'(r)}\,dr. \tag{3}$$

3.2 Conserved quantity and reduced branch

Because the reduced Lagrangian does not depend explicitly on v, there is a conserved quantity,

$$C = \frac{4\pi r^2\big(1 - f(r)v'(r)\big)}{\sqrt{-f(r)\,v'(r)^2 + 2v'(r)}}. \tag{4}$$

Defining

$$Q(r;C) = C^2 + (4\pi r^2)^2 f(r), \tag{5}$$

the smooth maximal-slice branch takes the compact form

$$v'(r) = \frac{1 - C/\sqrt{Q(r;C)}}{f(r)}, \tag{6}$$

while the volume integrand becomes

$$\frac{dV}{dr} = \frac{(4\pi r^2)^2}{\sqrt{Q(r;C)}}. \tag{7}$$

This reduction is the technical backbone of the whole program. Every model-specific result in the paper is obtained by specifying $f(r)$, determining the admissible branch, and locating the maximizing point of the induced critical quantity.

3.3 Critical constant

The admissibility condition is $Q(r;C) \geq 0$ on the physical interval $0 < r < r_+$. The critical constant is therefore

$$\boxed{C^2_{\text{crit}}(M, \text{deformation}) = \max_{0<r<r_+}\Big[-(4\pi r^2)^2 f(r)\Big].} \tag{8}$$

This quantity is the central invariant of the VIG framework.

4 Schwarzschild benchmark

For Schwarzschild,

$$f_{\mathrm{Schw}}(r) = 1 - \frac{2M}{r}.$$

The critical point is

$$r_* = \frac{3M}{2},$$

and the critical constant is exactly

$$\boxed{C_{\mathrm{crit}}(M,0) = 3\sqrt{3}\,\pi M^2.} \tag{9}$$

This is precisely the Christodoulou–Rovelli growth coefficient [1]. It serves as the anchor of the comparative theory: every regular model studied here reduces continuously to this value at zero deformation.

5 Hayward model

5.1 Metric and exact reduction

For Hayward,

$$f_{\mathrm{Hay}}(r) = 1 - \frac{2Mr^2}{r^3 + 2M\ell^2}. \tag{10}$$

Introduce the dimensionless variables

$$x = \frac{r}{M}, \qquad \lambda = \frac{\ell}{M}.$$

Then

$$f_{\mathrm{Hay}}(x,\lambda) = 1 - \frac{2x^2}{x^3 + 2\lambda^2}. \tag{11}$$

Defining

$$\Phi_\lambda(x) := -x^4 f_{\mathrm{Hay}}(x,\lambda) = \frac{x^4\left(2x^2 - x^3 - 2\lambda^2\right)}{x^3 + 2\lambda^2}, \tag{12}$$

the exact Hayward critical law is

$$\boxed{C_{\mathrm{crit}}^{\mathrm{Hay}}(M,\ell) = 4\pi M^2\,\mathcal{C}_{\mathrm{Hay}}(\lambda), \qquad \mathcal{C}_{\mathrm{Hay}}(\lambda) = \sqrt{\max_{0<x<x_+} \Phi_\lambda(x)}.} \tag{13}$$

5.2 Critical and horizon equations

Let $x_*(\lambda)$ maximize $\Phi_\lambda(x)$. Then the critical point satisfies

$$\boxed{2x^6 - 3x^5 + 8\lambda^2 x^3 - 12\lambda^2 x^2 + 8\lambda^4 = 0.} \tag{14}$$

The outer horizon is determined by

$$\boxed{x_+^3 - 2x_+^2 + 2\lambda^2 = 0.} \tag{15}$$

At extremality,

$$\boxed{\lambda_c = \frac{4}{3\sqrt{3}}, \qquad x_c = \frac{4}{3}.} \tag{16}$$

5.3 Hayward asymptotics

Near $\lambda = 0$,

$$x_*(\lambda) = \frac{3}{2} - \frac{128}{243}\lambda^4 + \mathcal{O}(\lambda^6), \tag{17}$$

and

$$\boxed{C_{\text{crit}}^{\text{Hay}}(M,\ell) = 3\sqrt{3}\,\pi M^2 \left[1 - \frac{32}{27}\left(\frac{\ell}{M}\right)^2 + \mathcal{O}\left(\frac{\ell^4}{M^4}\right)\right].} \tag{18}$$

Near extremality, with

$$t = \lambda_c^2 - \lambda^2,$$

one finds

$$x_+(\lambda) = \frac{4}{3} + \sqrt{t} - \frac{1}{4}t + \frac{5}{32}t^{3/2} + \mathcal{O}(t^2), \tag{19}$$

$$x_*(\lambda) = \frac{4}{3} + \frac{3}{4}t - \frac{135}{64}t^2 + \mathcal{O}(t^3), \tag{20}$$

and the critical-growth law becomes

$$\boxed{C_{\text{crit}}^{\text{Hay}}(M,\ell) = \frac{16\pi}{3}M^2\sqrt{t} + 3\pi M^2 t^{3/2} - \frac{99\pi}{32}M^2 t^{5/2} + \mathcal{O}(t^{7/2}).} \tag{21}$$

5.4 Hayward numerical status

The exact Hayward analytic law was checked directly against the numerical scan products exported by the Hayward unified framework. The source products `vig_interior_atlas.csv`, `vig_interior_atlas_unified` and `vig_unified_master_summary.csv` supply the atlas, summary, and asymptotic-validation layers. In the master results, Hayward's tail-to-critical ratio averages approximately

$$\frac{\text{tail mean}}{C_{\text{crit}}^{\text{Hay}}} \approx 1.001206,$$

showing that the tail law is not merely qualitative but numerically tight.

6 Bardeen model

6.1 Metric and exact reduction

For Bardeen,

$$f_{\text{Bar}}(r) = 1 - \frac{2Mr^2}{(r^2+g^2)^{3/2}}. \tag{22}$$

Introduce

$$x = \frac{r}{M}, \qquad \beta = \frac{g}{M}.$$

Then

$$f_{\text{Bar}}(x,\beta) = 1 - \frac{2x^2}{(x^2+\beta^2)^{3/2}}. \tag{23}$$

Defining

$$\Phi_\beta(x) := -x^4 f_{\text{Bar}}(x,\beta) = \frac{x^4\big(2x^2 - (x^2+\beta^2)^{3/2}\big)}{(x^2+\beta^2)^{3/2}}, \tag{24}$$

the exact Bardeen critical law is

$$\boxed{C_{\rm crit}^{\rm Bar}(M,g) = 4\pi M^2\,\mathcal{C}_{\rm Bar}(\beta), \qquad \mathcal{C}_{\rm Bar}(\beta) = \sqrt{\max_{0<x<x_+}\Phi_\beta(x)}.} \tag{25}$$

6.2 Auxiliary variable and exact equations

For Bardeen, the reduction becomes especially transparent in the auxiliary variable

$$y = \sqrt{x^2+\beta^2}.$$

Then $x^2 = y^2 - \beta^2$, and the critical-point equation collapses to

$$\boxed{2y^5 - 3y^4 + 3\beta^4 = 0.} \tag{26}$$

The horizon equation becomes

$$\boxed{y^3 - 2y^2 + 2\beta^2 = 0.} \tag{27}$$

Thus the exact Bardeen law is obtained by solving (26) for the physical root $y_*(\beta)$, reconstructing $x_*(\beta) = \sqrt{y_*(\beta)^2 - \beta^2}$, and evaluating

$$C_{\rm crit}^{\rm Bar}(M,g) = 4\pi M^2\, x_*(\beta)^2\sqrt{-f_{\rm Bar}(x_*(\beta),\beta)}.$$

6.3 Bardeen extremality and asymptotics

The extremal threshold is

$$\boxed{\beta_c = \frac{4}{3\sqrt{3}}.} \tag{28}$$

At the extremal horizon,

$$y_c = \frac{4}{3}, \qquad x_{+,c}^2 = \frac{32}{27}, \qquad x_{+,c} = \frac{4\sqrt{6}}{9}.$$

Near $\beta = 0$,

$$y_*(\beta) = \frac{3}{2} - \frac{8}{27}\beta^4 + \mathcal{O}(\beta^8), \tag{29}$$

and

$$\boxed{C_{\rm crit}^{\rm Bar}(M,g) = 3\sqrt{3}\,\pi M^2\left[1 - \frac{4}{3}\left(\frac{g}{M}\right)^2 + \mathcal{O}\left(\frac{g^4}{M^4}\right)\right].} \tag{30}$$

Near extremality, with

$$t = \beta_c^2 - \beta^2,$$

one finds

$$y_+(\beta) = \frac{4}{3} + \sqrt{t} - \frac{1}{4}t + \frac{5}{32}t^{3/2} + \mathcal{O}(t^2), \tag{31}$$

$$y_*(\beta) = \frac{4}{3} + \frac{9}{8}t - \frac{3645}{512}t^2 + \mathcal{O}(t^3), \tag{32}$$

and the leading critical-growth law becomes

$$\boxed{C_{\rm crit}^{\rm Bar}(M,g) = \frac{16\sqrt{6}\,\pi}{9}M^2\sqrt{t} + \frac{21\sqrt{6}\,\pi}{8}M^2 t^{3/2} - \frac{13689\sqrt{6}\,\pi}{2048}M^2 t^{5/2} + \mathcal{O}(t^{7/2}).} \tag{33}$$

6.4 Bardeen numerical status

The exact Bardeen law is verified by the source products `vig_bardeen_atlas.csv`, `vig_bardeen_master_summary.c` `vig_bardeen_edge_atlas.csv`, and `vig_bardeen_edge_master_summary.csv`. The dense edge scan shows that the near-extremal expansion enters its expected regime near the edge and that the tail law remains numerically tight throughout. In the master results, the mean Bardeen tail-to-critical ratio is approximately

$$\frac{\text{tail mean}}{C_{\text{crit}}^{\text{Bar}}} \approx 1.001258.$$

The Hayward and Bardeen tail laws are therefore comparably stable.

7 Comparative full-edge results

7.1 Universal structure

The following pattern now appears in every model studied:

Proposition 7.1 (Comparative VIG universality, empirical form)**.** *For Schwarzschild, Hayward, and Bardeen, the near-critical maximal-slice family obeys*

$$V(v) \sim C_{\text{crit}}^{(\text{model})}(M,u)\, v,$$

with the numerical tail satisfying

$$\textit{tail mean} \approx C_{\text{crit}}^{(\text{model})}.$$

The Schwarzschild anchor is recovered at vanishing deformation:

$$C_{\text{crit}}^{(\text{model})}(M,0) = 3\sqrt{3}\,\pi M^2.$$

The common qualitative geometry is equally clear:

- the outer horizon shrinks as deformation grows,
- the critical slice radius also shrinks,
- the admissible maximal-slice channel compresses,
- the normalized critical-growth law decreases monotonically.

7.2 Model-specific structure

The universality does *not* mean identical deformation curves. At equal normalized deformation u,

$$\frac{C_{\text{crit}}^{\text{Bar}}}{3\sqrt{3}\,\pi M^2} < \frac{C_{\text{crit}}^{\text{Hay}}}{3\sqrt{3}\,\pi M^2} \qquad (u > 0) \tag{34}$$

through the full overlapping comparison range available in the master outputs.

The decisive new result is the full-edge comparison. Using the aggregated products `master_full_edge_common_g` and `master_full_edge_combined_atlas.csv`, the comparison now extends to the top of the overlapping normalized interval. The resulting behavior is unambiguous:

- At $u = 0$, the two models coincide, as required by the Schwarzschild anchor.

- As u increases, Bardeen remains below Hayward at every sampled point in the normalized critical-growth law.
- The difference is mild at small deformation, but accelerates strongly near the edge.
- By the top of the overlap, the ratio has fallen to approximately

$$\frac{C_{\text{crit}}^{\text{Bar}}/(3\sqrt{3}\pi M^2)}{C_{\text{crit}}^{\text{Hay}}/(3\sqrt{3}\pi M^2)} \approx 0.837,$$

  so Bardeen's normalized critical-growth law is about 16.3% lower than Hayward's there.

This goes well beyond the earlier partial-overlap comparison. The two models remain in the same qualitative class, but they do not remain quantitatively close near the edge.

Remark 7.2. *The weak-deformation asymptotics already hint at the correct hierarchy:*

$$C_{\text{crit}}^{\text{Hay}} = 3\sqrt{3}\,\pi M^2 \left[1 - \frac{32}{27}u^2 + \cdots\right], \qquad C_{\text{crit}}^{\text{Bar}} = 3\sqrt{3}\,\pi M^2 \left[1 - \frac{4}{3}u^2 + \cdots\right].$$

Since $\frac{4}{3} > \frac{32}{27}$*, Bardeen bends downward faster near* $u = 0$*. The full-edge comparison shows that this early tendency amplifies rather than washes out.*

7.3 Geometric comparison

The geometric comparison is just as important as the growth-law comparison.

- Bardeen has smaller x_+ than Hayward at equal normalized deformation.
- Bardeen has smaller x_* than Hayward at equal normalized deformation.
- The entire Bardeen structure is therefore pushed inward more strongly.

The most interesting twist lies in the interior gap

$$x_+ - x_*.$$

Although both x_+ and x_* are smaller in Bardeen, the comparative plots show that the Bardeen gap is slightly *larger* than the Hayward gap over the overlap region. So Bardeen is not merely applying a uniform rescaling of the Hayward geometry. It is pushing the whole structure inward more strongly while preserving a slightly wider separation between the outer horizon and the critical slice.

This makes the comparative result geometrically vivid: the two models share the same type of interior growth law, but the way they deform the interior channel is not the same.

8 Reproducibility architecture

Version 7 Final is intended to be reproducible from a finite and explicit code–data chain.

8.1 Source codes

The core generating codes are:

File	Role
vig_unified_framework_v4.py	Hayward unified framework; generates Hayward atlas and Hayward master summaries.
vig_bardeen_unified_v1.py	Bardeen unified framework; generates Bardeen atlas and summary products.
vig_bardeen_edge_scan_v1.py	Dense Bardeen edge scan and asymptotic validation near extremality.
vig_hayward_bardeen_full_edge_comparison.py	Full-edge comparative overlay between Hayward and Bardeen.
vig_master_driver.py	Top-level aggregation layer; exports master CSVs and final figure suite.
universal_clock.py	Original clock bookkeeping and invariant summary.

8.2 Source data products

The principal source CSVs are:

CSV product	Role
vig_analytic_ccrit_validation.csv	Hayward analytic/numeric validation layer.
vig_interior_atlas.csv	Hayward atlas.
vig_interior_atlas_unified.csv	Hayward unified atlas export.
vig_unified_master_summary.csv	Hayward unified master summary.
vig_bardeen_atlas.csv	Bardeen atlas.
vig_bardeen_master_summary.csv	Bardeen summary layer.
vig_bardeen_edge_atlas.csv	Dense Bardeen edge atlas.
vig_bardeen_edge_master_summary.csv	Dense Bardeen edge summary layer.
vig_hayward_bardeen_full_edge_comparison.csv	Combined comparison export.
vig_hayward_bardeen_full_edge_common_grid.csv	Common interpolation grid for full-edge comparison.

8.3 Master outputs

The master driver compiles these layers into the final aggregation products:

Master output	Role
master_universal_clock_summary.csv	Clock bookkeeping summary.
master_hayward_atlas.csv	Final Hayward atlas used for cross-model comparison.
master_bardeen_atlas.csv	Final Bardeen atlas used for cross-model comparison.
master_full_edge_common_grid.csv	Canonical common grid for the full-edge overlap.
master_full_edge_combined_atlas.csv	Combined model atlas for comparison.
master_multimodel_summary.csv	Compact summary of model ranges and tail-tracking statistics.
master_manifest.csv	Artifact manifest for the aggregated run.

The manuscript figures are tied directly to the corresponding exported PNGs from this pipeline. In other words, the final paper is not merely inspired by the code; it is derived from an explicit code path and a finite set of named outputs.

9 Selected numerical outputs

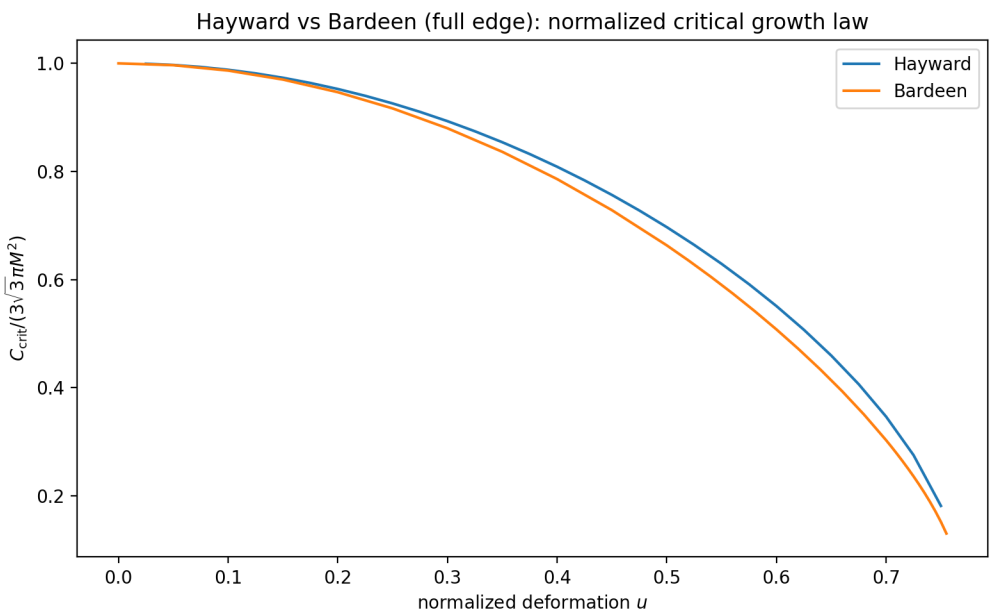

Figure 1: Hayward vs. Bardeen overlay of the normalized critical-growth law over the full overlapping normalized-deformation range.

10 Discussion and scope

The central accomplishment of the present program is not the derivation of a universal saturation time. It is the discovery and verification of a comparative geometric law for regular interiors. The

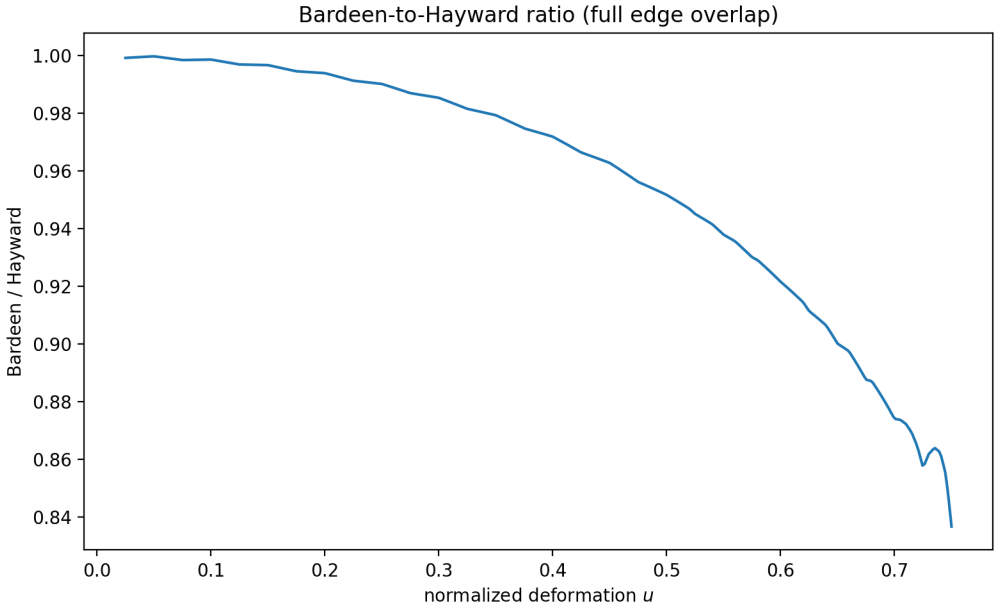

Figure 2: Bardeen-to-Hayward ratio of the normalized critical-growth law over the full-edge overlap. The ratio falls from the Schwarzschild anchor to approximately 0.837 near the top of the overlap.

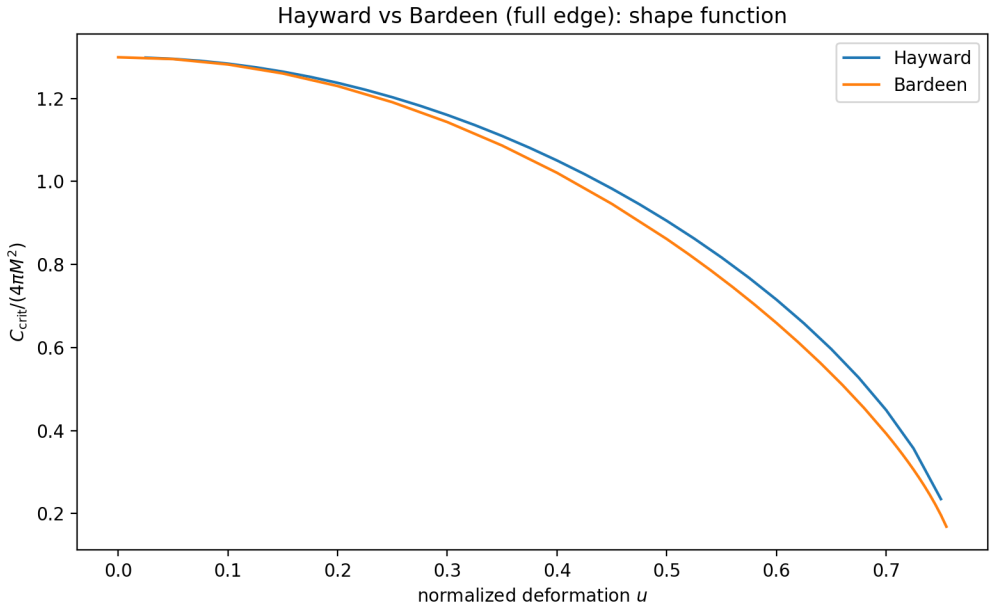

Figure 3: Comparison of the shape function $C_{\mathrm{crit}}/(4\pi M^2)$ for Hayward and Bardeen. The separation is not a normalization artifact; it is present directly in the underlying dimensionless law.

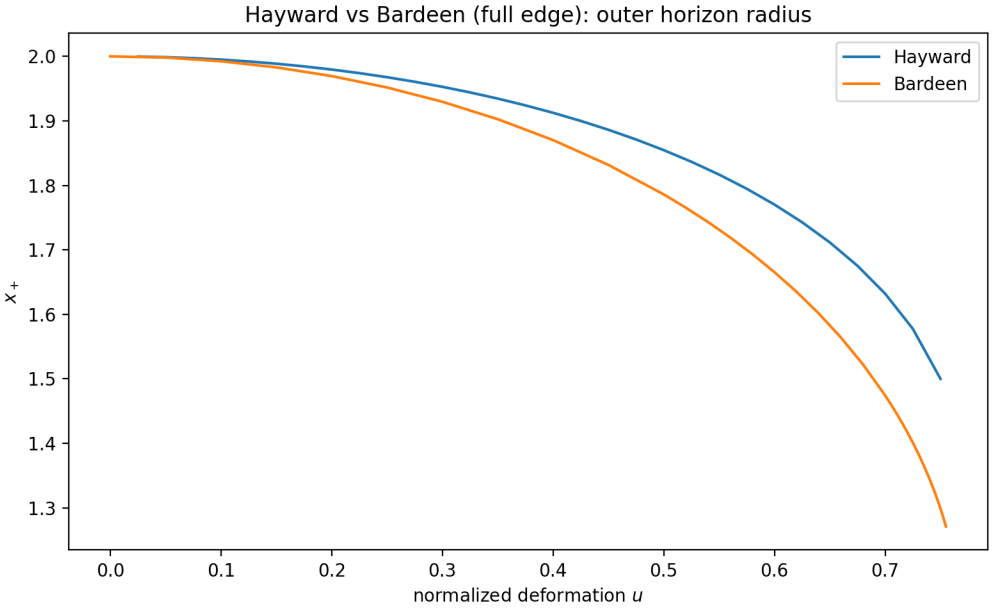

Figure 4: Outer-horizon comparison between Hayward and Bardeen over the full-edge overlap. Bardeen contracts the outer horizon more strongly.

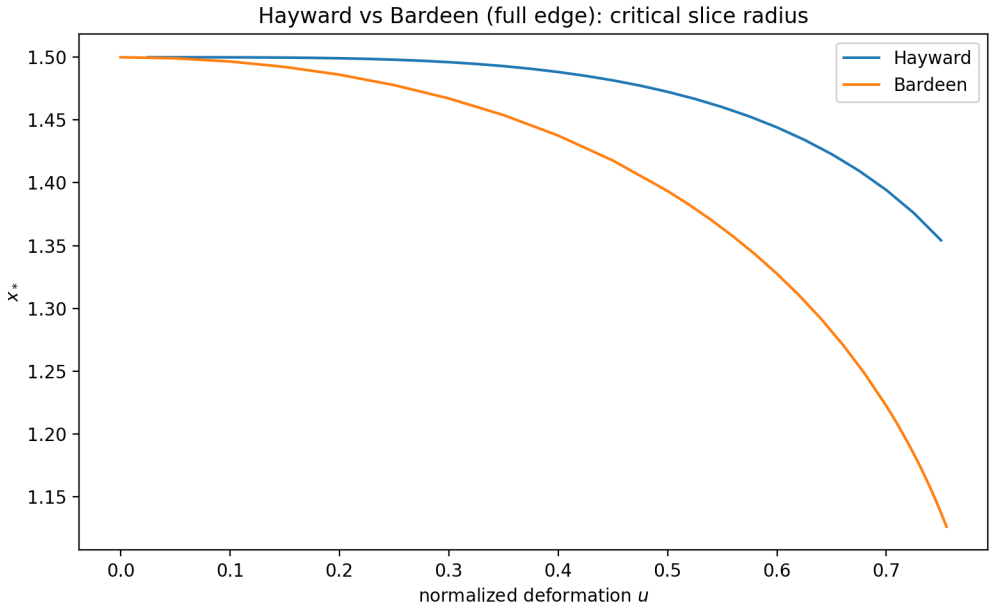

Figure 5: Critical-slice comparison between Hayward and Bardeen over the full-edge overlap. Bardeen pushes the critical slice inward more strongly.

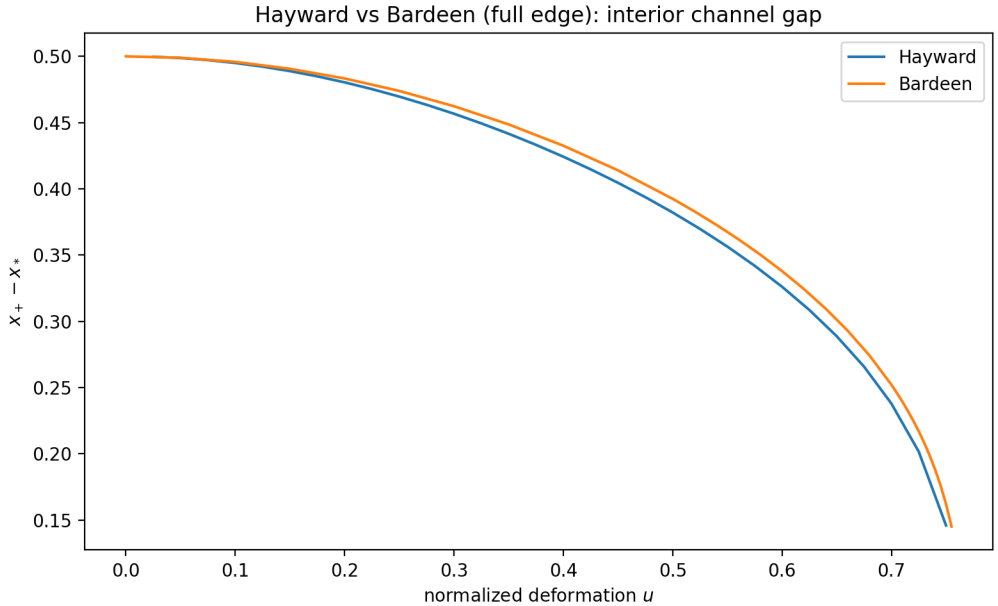

Figure 6: Comparison of the interior channel gap $x_+ - x_*$. Although Bardeen has smaller x_+ and smaller x_*, its gap is slightly larger over the overlap.

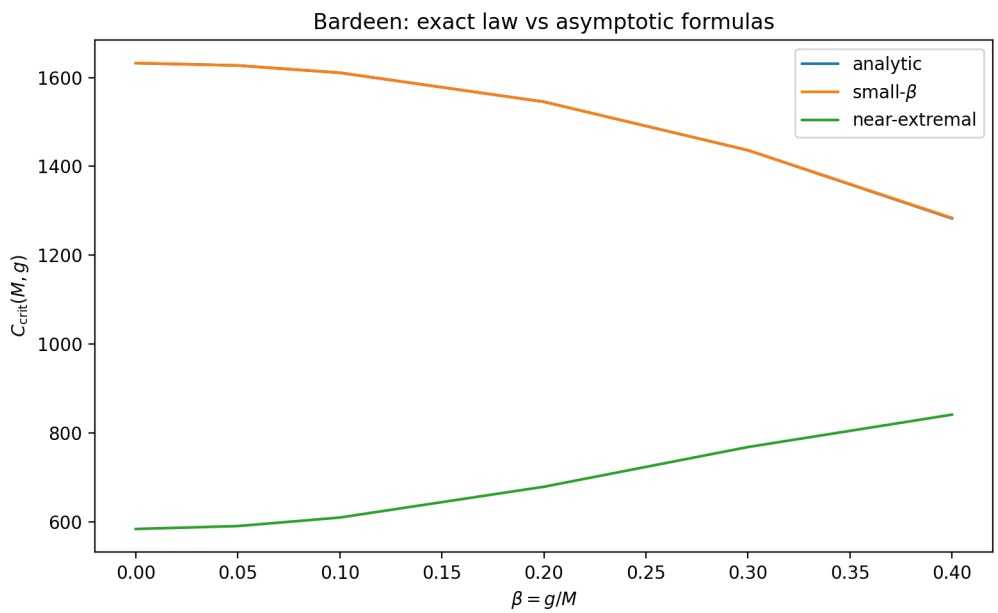

Figure 7: Bardeen exact law versus weak-deformation and near-extremal asymptotics in the dense edge scan.

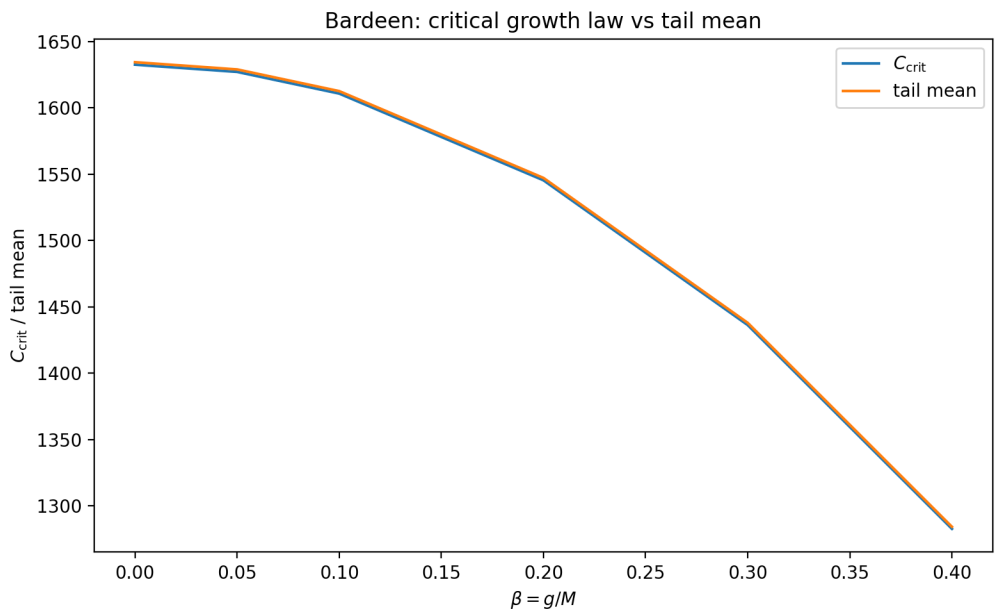

Figure 8: Bardeen edge scan: the near-critical family tail continues to track $C_{\rm crit}$ toward extremality.

strongest validated statement is

$$V(v) \sim C_{\rm crit}^{(\rm model)}(M,u)\,v, \qquad \text{tail mean} \approx C_{\rm crit}^{(\rm model)}.$$

That statement is supported by exact reduction, asymptotic analysis, family scans, cross-model comparison, and a master reproducibility pipeline.

The original clock program remains conceptually alive but formally subordinate. Any future argument for a genuine saturation clock must now explain how it emerges from, modifies, or ultimately replaces the validated geometric critical-growth law. In that sense, the clock conjecture has become a *downstream* problem rather than the primary theorem candidate.

The comparative outcome is also physically suggestive. Hayward and Bardeen both regularize the black-hole interior, but they do so differently enough that the resulting interior cartographies visibly diverge near the edge. The shared class is real; the model-specific deformation is also real. This is exactly the kind of result a comparative program should produce.

11 Conclusion

Version 7 Final is the polished culmination of the current VIG program.

The main result is no longer a one-model speculation about a possible interior clock. It is a two-model universality statement about near-critical maximal-slice growth:

$$\boxed{V(v) \sim C_{\rm crit}^{(\rm model)}(M,u)\,v, \qquad \text{tail mean} \approx C_{\rm crit}^{(\rm model)}.}$$

What is universal:

- Schwarzschild anchoring at zero deformation,
- monotone suppression of the critical-growth law,

- compression of the admissible interior channel,
- numerical tracking of the tail by the model-specific $C_{\rm crit}$.

What is model-specific:

- the exact deformation curve $\mathcal{C}(u)$,
- the detailed horizon and critical-slice radii,
- the strength of suppression at fixed normalized deformation,
- the edge behavior of the comparative ratio.

Hayward and Bardeen therefore belong to the same qualitative interior-growth class, while Bardeen deforms that class more strongly. The full-edge comparison shows that this is not a small perturbative effect confined to weak deformation; it becomes substantial near the overlap boundary and is visible simultaneously in the growth law, the shape function, and the horizon–slice geometry. This is the strongest, cleanest, and most reproducible statement the project has produced so far.

A Collected exact laws

A.1 Hayward

$$C_{\rm crit}^{\rm Hay}(M,\ell) = 4\pi M^2\,\mathcal{C}_{\rm Hay}(\lambda), \qquad \lambda = \frac{\ell}{M}.$$

$$2x^6 - 3x^5 + 8\lambda^2 x^3 - 12\lambda^2 x^2 + 8\lambda^4 = 0, \qquad x_+^3 - 2x_+^2 + 2\lambda^2 = 0.$$

A.2 Bardeen

$$C_{\rm crit}^{\rm Bar}(M,g) = 4\pi M^2\,\mathcal{C}_{\rm Bar}(\beta), \qquad \beta = \frac{g}{M}.$$

$$2y^5 - 3y^4 + 3\beta^4 = 0, \qquad y^3 - 2y^2 + 2\beta^2 = 0.$$

B Collected weak-deformation laws

$$C_{\rm crit}^{\rm Hay}(M,\ell) = 3\sqrt{3}\,\pi M^2\left[1 - \frac{32}{27}\left(\frac{\ell}{M}\right)^2 + \cdots\right]$$

$$C_{\rm crit}^{\rm Bar}(M,g) = 3\sqrt{3}\,\pi M^2\left[1 - \frac{4}{3}\left(\frac{g}{M}\right)^2 + \cdots\right]$$

C Collected near-extremal laws

For Hayward,

$$C_{\rm crit}^{\rm Hay}(M,\ell) = \frac{16\pi}{3}M^2\sqrt{t} + 3\pi M^2 t^{3/2} - \frac{99\pi}{32}M^2 t^{5/2} + \cdots, \qquad t = \lambda_c^2 - \lambda^2.$$

For Bardeen,

$$C_{\rm crit}^{\rm Bar}(M,g) = \frac{16\sqrt{6}\,\pi}{9}M^2\sqrt{t} + \frac{21\sqrt{6}\,\pi}{8}M^2 t^{3/2} - \frac{13689\sqrt{6}\,\pi}{2048}M^2 t^{5/2} + \cdots, \qquad t = \beta_c^2 - \beta^2.$$

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